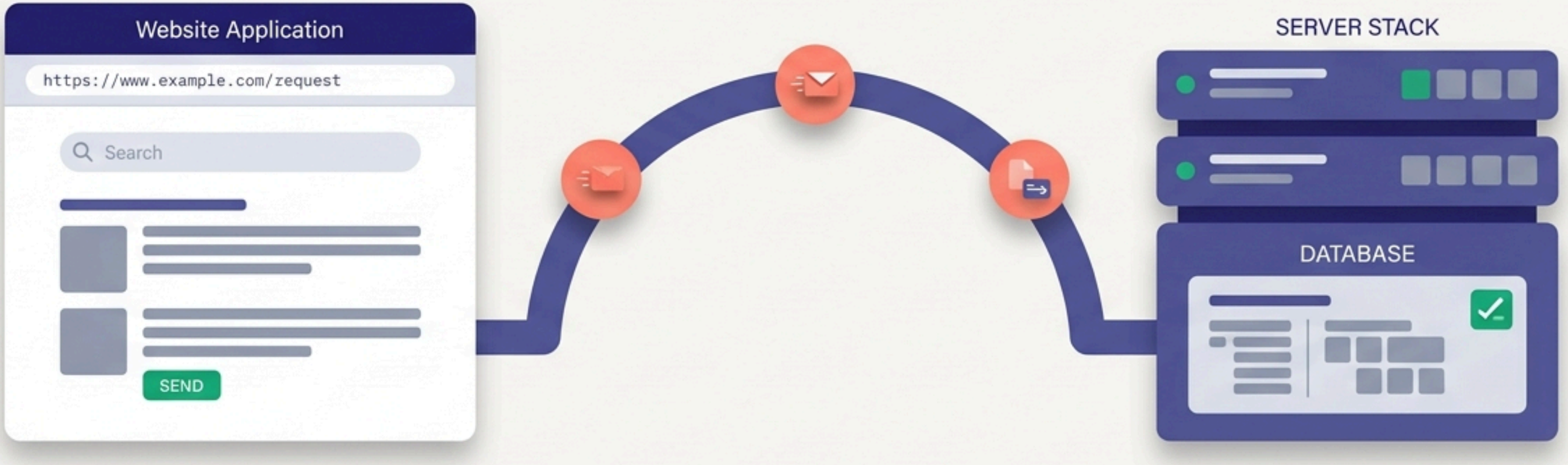
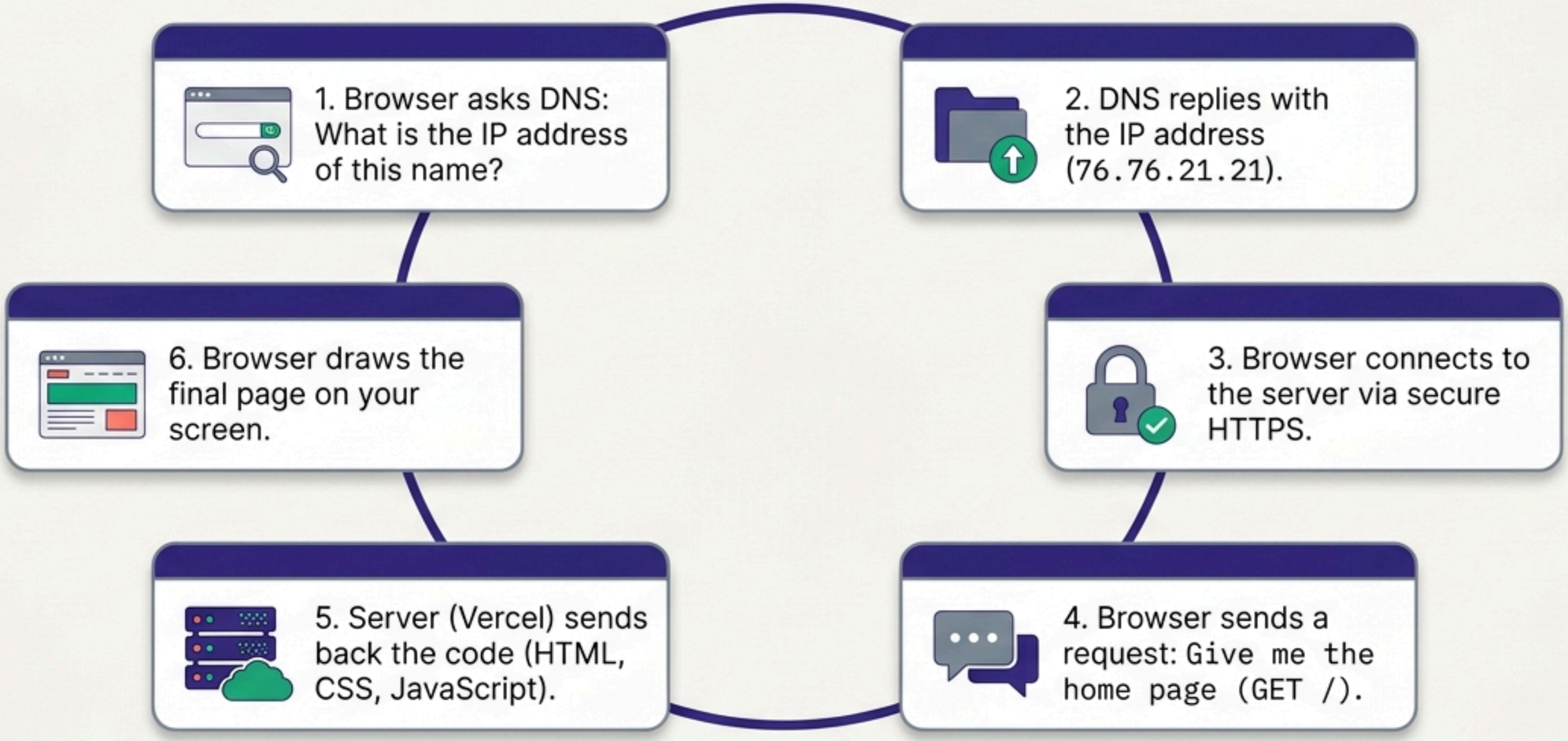
Anatomy of a Digital Conversation

How the Web Works: From Click to Code

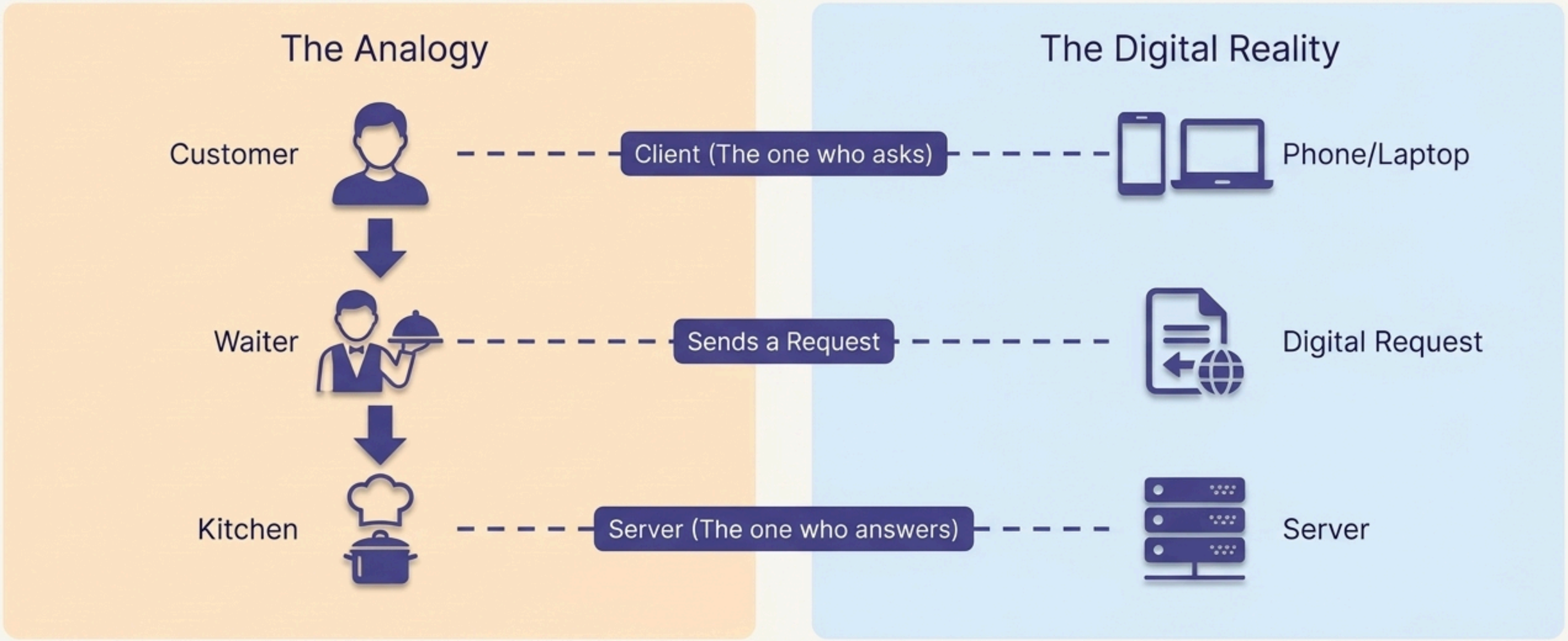


The Macro Journey

The Action: You type `campusbite.vercel.app` and press Enter. (Happens in under a second).



Two Actors: Client & Server



Key Detail: One server can simultaneously serve thousands of clients.

How They Find Each Other: DNS

Domain:
Humans read words.

`campusbite.vercel.app`



IP Address:
Machines read numbers.

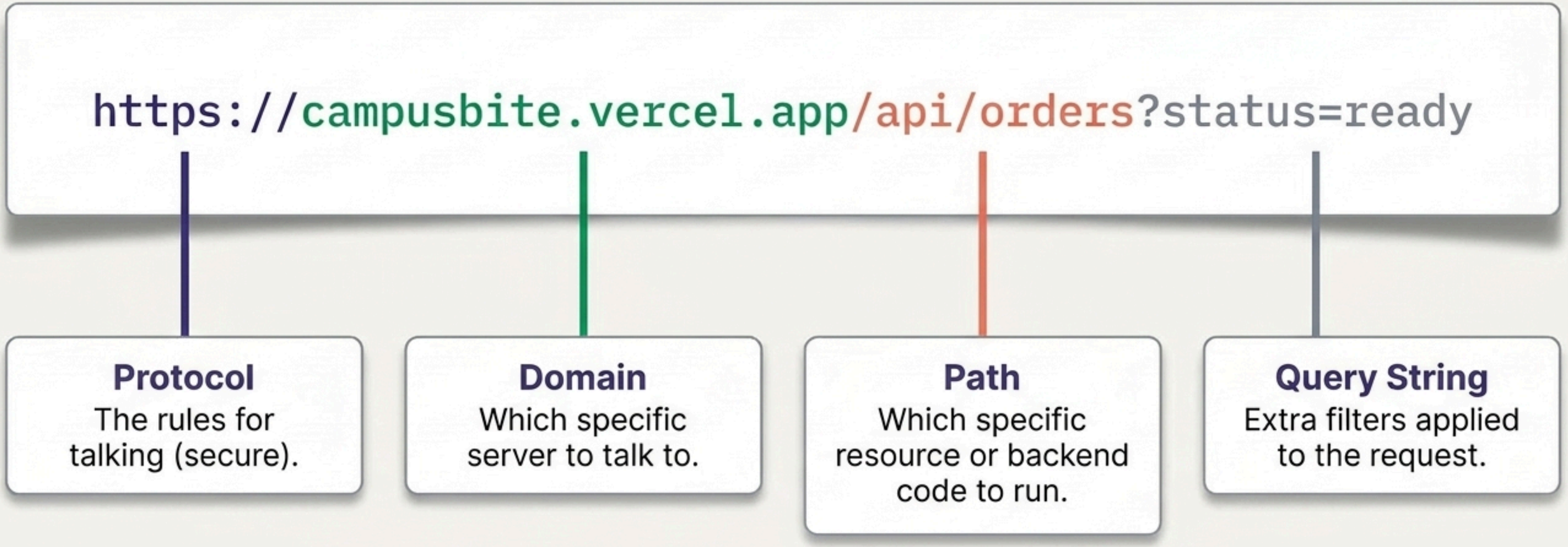
`76.76.21.21`

Domain
The easy-to-remember contact name.

IP Address
The real numeric address of the server.

DNS
The system that maps the domain to the IP address.

The Anatomy of a URL



Key Detail: A URL is the full address of one specific resource.

HTTP: The Digital Envelope

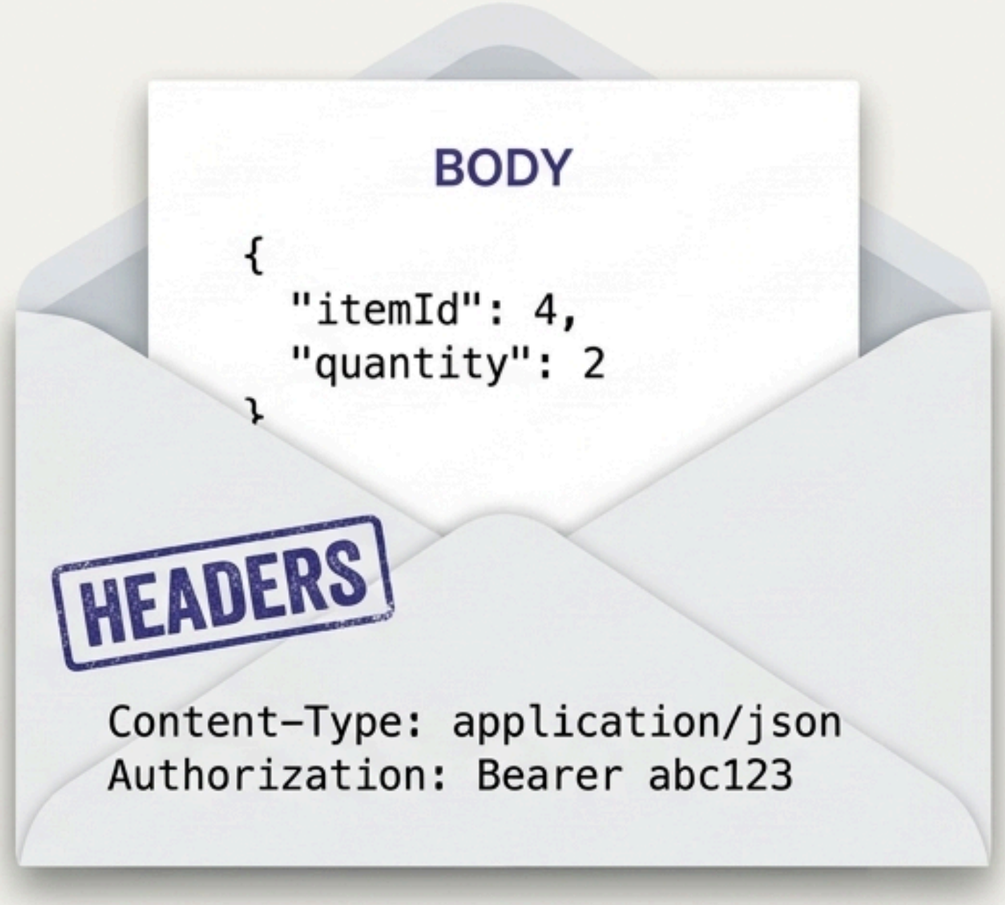
The language browsers and servers use to talk.

The Outside (Headers)

Like the address on an envelope. Includes metadata and security passes.

The Inside (Body)

Like the letter inside. The actual data being sent.



Every Request: Method, URL, Headers, (Optional) Body.

Every Response: Status Code, Headers, (Optional) Body.

The CRUD Action Matrix: HTTP Methods

Method	Core Action	CRUD Mapping	CampusBite Example
GET	Read data (“Show me”)	Read	`Get the menu: GET /api/menu`
POST	Create new data (“Add this”)	Create	`Place an order: POST /api/orders`
PUT / PATCH	Update existing data (“Change this”)	Update	`Mark order ready: PATCH /api/orders/1021`
DELETE	Remove data (“Remove this”)	Delete	`Cancel an order: DELETE /api/orders/1021`

The Verdict Matrix: Status Codes



2xx - Success ('All good')

- `200 OK`: It worked (Menu loaded).
- `201 Created`: New item created (Order placed).



4xx - Client Error ('You made a mistake')

- `400 Bad Request`: You sent wrong data (Quantity is -3).
- `401 Unauthorized`: Not logged in / wrong passcode.
- `404 Not Found`: That thing does not exist (Order 99999).



5xx - Server Error ('We made a mistake')

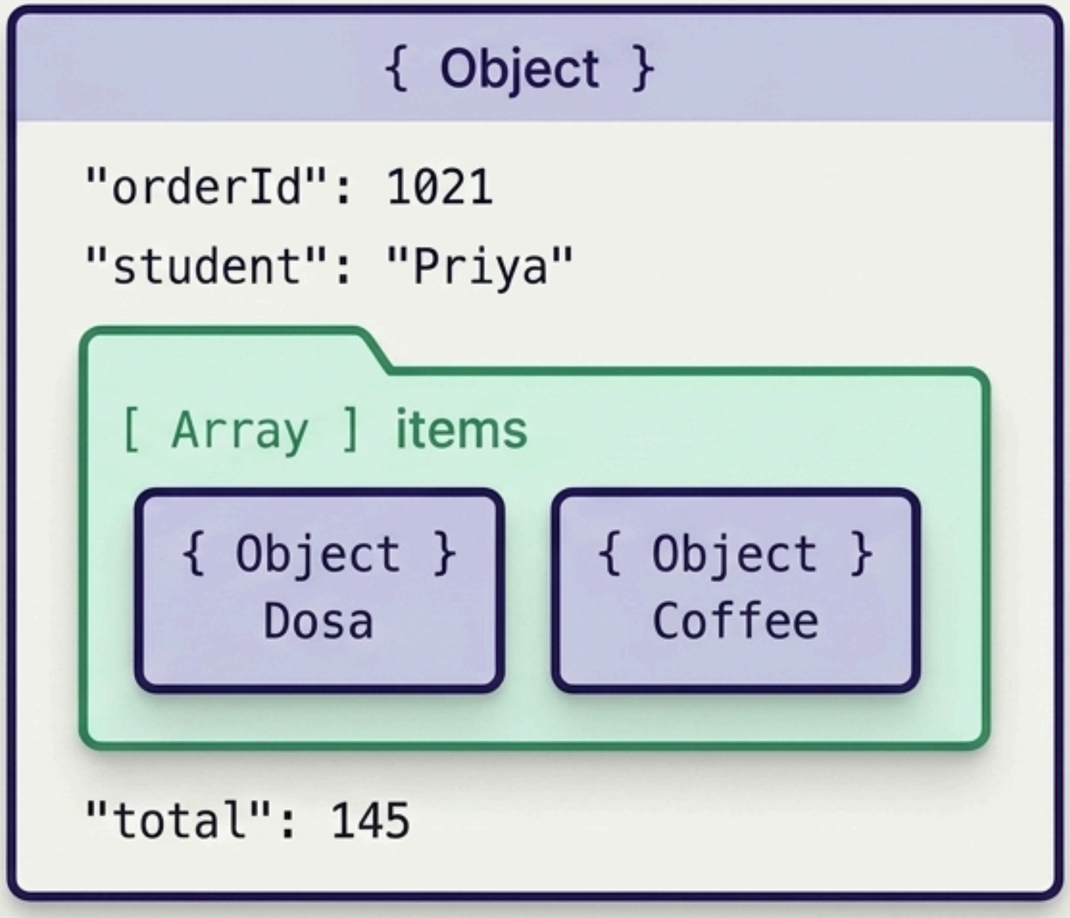
- `500 Server Error`: Our code broke (Bug in the backend).

Debugging Tip: Check 4xx on the request (your code), check 5xx in the server logs (backend code).

Unpacking the Cargo: What is JSON?

```
{
  "orderId": 1021,
  "student": "Priya",
  "items": [
    { "name": "Dosa", "quantity": 2 },
    { "name": "Coffee", "quantity": 1 }
  ],
  "total": 145
}
```

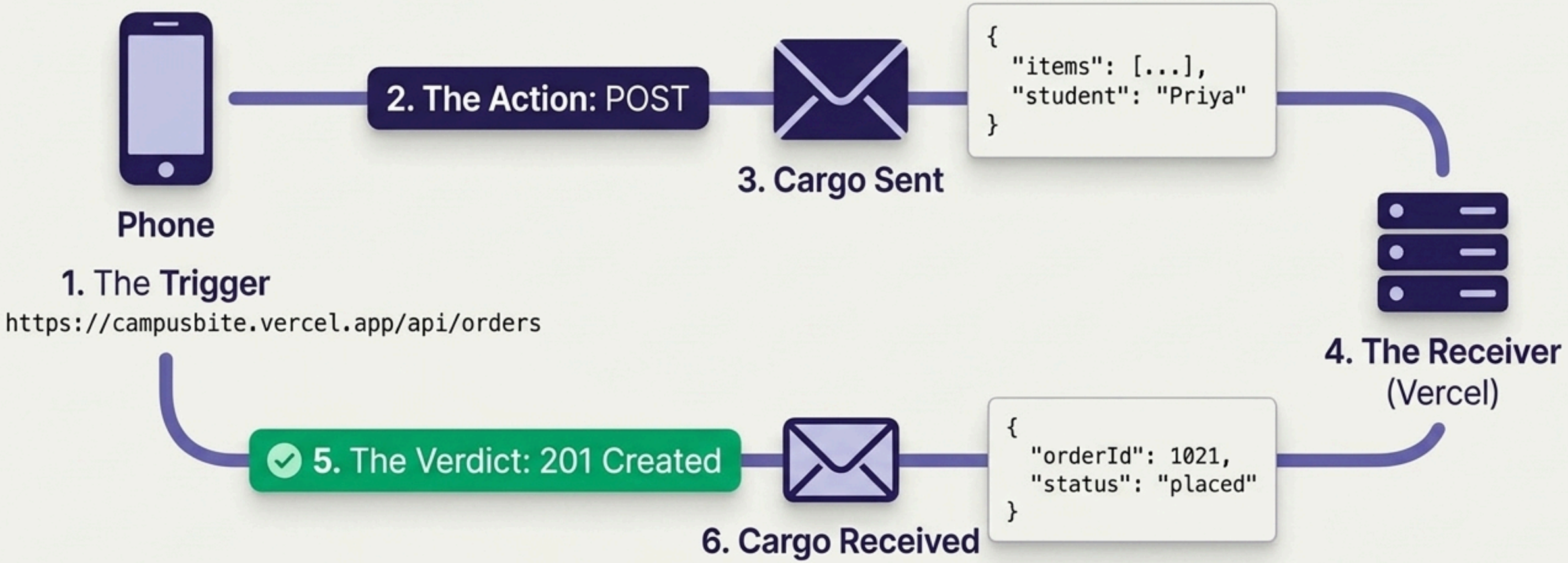
- JSON (JavaScript Object Notation): A simple text format for sending data.
- Readable by every language and AI model.



{ } = **Object** (thing with properties)
[ ] = **Array** (list)

Synthesis: A Complete Digital Conversation

Placing an Order on CampusBite



Takeaway: A seamless digital conversation, executed in milliseconds, utilizing DNS, HTTP, CRUD, and JSON.